



Formation mechanisms of intra-organizational membership overlap: a longitudinal network analysis of membership data from the International Communication Association

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Abstract

Social entities are linked and interdependent through shared members, a phenomenon described as niche overlap. Using archival data on yearly affiliations of International Communication Association (ICA) members with divisions and interest groups over a 9-year period, this study conducts a longitudinal network analysis to examine the formation mechanisms of intra-organizational niche overlap among organizational subunits. The results of the temporal exponential random graph model suggest the absence of dominant ICA subunits whose topic areas are broad enough to integrate different specializations through membership overlap. Triadic closure facilitates the sharing of members among three subunits, leading to a cohesive internal structure and limited brokerage opportunities. Subunits with smaller membership sizes are more likely to have membership overlap, particularly with larger subunits. Positive affinity relations between subunits, characterized by the sharing of scarce resources within the organization, contribute to the formation of membership overlap.

Keywords: niche overlap, ecology, network analysis, intra-organizational communication, ICA

Membership overlap occurs when individuals belong to the same set of groups, teams, or organizations. Although the current information-rich environment has raised concerns about fragmentation, wherein social entities share very few or even no members, empirical research has shown the prevalence of membership overlap in physical and virtual spaces (Webster, 2014; Webster & Ksiazek, 2012). This pervasiveness is attributed to permeable category boundaries, reduced participation costs, flexible membership options, and the rise of networked individualism (Wellman et al., 2003). Although membership overlap leads to mutually beneficial relationships by creating pathways for information flows and resource sharing, it makes social entities vulnerable to competition because an individual's time or behavioral commitment is a limited and scarce resource (Wang et al., 2013). As a communication signal, the extensive membership overlap between entities also dilutes the distinctiveness of each, creating potential challenges for the development of focused and exclusive in-group identity (Jetten et al., 1998).

This article aims to integrate ecological and network approaches to communication (Lee & Monge, 2011; Monge & Poole, 2008) by exploring the formation mechanisms of intra-organizational membership overlap, an important form of niche overlap. The incorporation of a communication network perspective into organizational ecology contributes to developing a middle-range niche theory that bridges between high levels of abstraction (e.g., ecological terms such as niche overlap) and empirical observations. The changing membership overlap between organizational segments is represented as a longitudinal network where links indicate their degree of overlap. This study

examines intra-organizational membership overlap in the context of the International Communication Association (ICA), which allows for a tailored membership experience through individual affiliations with subunits (i.e., divisions or interest groups) defined in terms of topic areas. Membership overlap within ICA arises when an individual holds membership in at least two subunits (Barnett & Danowski, 1992; Chung et al., 2009). The results of a longitudinal network analysis show that the creation of membership overlap is driven by the attributes and structural embeddedness of single subunits and subunit pairs or triads. The findings provide new insights into (a) the phenomenon of niche overlap within organizations and (b) the internal structure of ICA and the communication field.

The article is organized as follows. The literature review section starts with an ecological conceptualization and a network operationalization of membership overlap. Subsequent discussions explore the formation mechanisms of intra-organizational membership overlap, focusing on the roles of network concentration, triadic closure, size dependence, and network multiplexity. Following the method and results sections, the article concludes with a discussion of empirical findings and their implications.

Literature review

Ecology of affiliation and membership overlap

Organizational ecology acknowledges that social entities do not exist in isolation; instead, they are embedded in an ecosystem where diverse entities develop interdependent relationships with

each other through dynamic and complex interactions (Monge & Poole, 2008). Ecological interdependence between social entities is often examined as network ties, which act as a communication infrastructure for resource acquisition in the environment (Lee & Monge, 2011; Monge & Poole, 2008). Negative interdependence occurs through competition for finite resources, whereas positive interdependence (e.g., mutualism and symbiosis) emphasizes that social entities benefit from coexistence (Aldrich et al., 2020). This ecological account of organizational survival in the communication landscape is applicable across multiple levels of analysis, including intra-organizational, inter-organizational, and community ecologies (Weber, 2017; Yang & Taylor, 2015). At the intra-organizational level, an organizational segment (e.g., a division, department, unit, and subunit) is part of a system where multiple segments develop interdependent relationships. Just as members of animal species obtain resources (e.g., food, water, and habitat) from a biological ecosystem, the survival of the segment depends on its ability to access environmental resources (e.g., member time and effort, attention, and funding).

One line of research in organizational ecology, which was originally developed to understand the competition of voluntary associations for members, is recognized as the ecology of affiliation (McPherson, 1983, 2004). Affiliations take various forms, such as being a member of a social group or a consumer of a media product. The theory proposes that social entities compete in a k -dimensional Blau space defined by socio-demographic characteristics such as age, gender, ethnicity, level of education, and income. Each characteristic constitutes a dimension of the k coordinates describing the niche space from which entities acquire resources (McPherson, 2004).

Social entities position themselves in the niche space (see Figure 1). *Niche breadth*, reflecting the heterogeneity or diversity of members affiliated with an entity, is the volume of the k -dimensional space occupied by the entity (McPherson, 1983). *Generalists* and *specialists* can be differentiated based on their niche breadth. Generalists occupy a wide range of niches for survival, while specialists focus on narrow areas (Hannan & Freeman, 1989). The phenomenon that organizations experience growth in membership size without expanding their niche breadth or transitioning from specialists into generalists is explained by the homophily principle (McPherson et al., 2001), or the tendency for individuals to associate with similar others. As recruitment processes involve recommendations and referrals from current organizational members, homophily in network recruitment may

translate into increased membership homogeneity or similarity, leading to the replication of specific combinations of socio-demographic characteristics.

Niche overlap between entities is the area they jointly occupy in the multidimensional space. The amount of niche overlap in the coordinate system is interpreted as the level of competition intensity between social entities (McPherson, 1983). Generalists often face competition with both generalists and specialists with overlapping interests. High niche overlap suggests that the involved entities rely on the same set of resources and target similar socio-demographic characteristics for survival. However, this conceptualization of niche overlap requires a precise knowledge of the niche breadth of each entity across k socio-demographic dimensions (e.g., the age range of an organization's members), which is often unavailable to researchers for measurement.

Another common conceptualization of niche overlap is *membership overlap*, which refers to an individual's affiliations with multiple groups, teams, or organizations (Wageman et al., 2012; Woo & Myers, 2020; Xu & Peng, 2023). Membership overlap occurs when various social entities have similar goals, or when individuals have diverse interests (Wang et al., 2013). In contrast to the traditional ecological model of affiliation (McPherson, 1983, 2004), research on membership overlap adopts a more people-centric perspective, recognizing that niche overlap reflects not only competition between social entities but also their positive interdependence. Specifically, individuals devote their time, attention, and energy to various social groups that meet personal needs. Membership in one social group can produce a spillover effect, enhancing the experience of being a member of another group. The positive interdependence between groups highlights that social affiliations are mutually constitutive.

This study investigates the mechanisms that drive the creation of membership overlap in the intra-organizational ecology of ICA, where *organizational subunits* cooperate and compete to obtain limited resources such as audience attention, budget allocations, recognition, awards, and support from leadership. Organizational subunits are formal components of a larger organization designed to perform certain tasks and functions. From an ecological perspective, these subunits operate interdependently rather than in isolation. Despite its competitive implications, membership overlap facilitates communication between subunits and promotes the development of shared meanings and norms at the organizational level (Tasselli et al., 2020). In the context of ICA, organizational subunits consist primarily of divisions and interest groups labeled by ongoing or emerging topics of interest. The establishment of a new interest group involves filing a petition signed by 3% of existing members to the Board of Directors for review and approval. An interest group can change its status to a division when its membership rate in ICA is higher than 5% for two consecutive years. A division (or interest group) may be redesignated as an interest group (or be dissolved) when its membership rate falls below a certain threshold (International Communication Association, 2021).

Following the tradition of interpreting ecological interdependence as network ties (Monge & Poole, 2008) and analyzing overlap data as network data (Mukerjee et al., 2018), this study understands membership overlap among subunits as a longitudinal network. In the membership overlap network, nodes represent ICA divisions and interest groups, and edges are created using a filtering algorithm (Mukerjee et al., 2018). This

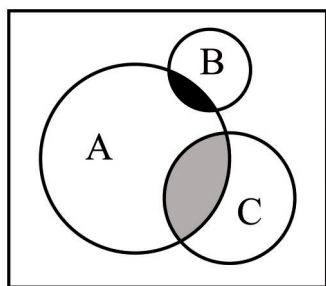


Figure 1. A graphic representation of the niche space, niche breadth, and niche overlap.

Note. The niche space is represented by a rectangular area. The three circles represent the niche breadths of entities A, B, and C. The darkly shaded area indicates the degree of niche overlap between A and B. The lightly shaded area indicates the degree of niche overlap between A and C.

algorithm determines whether the degree of overlap between node pairs significantly exceeds the threshold of connection strength. Edges are present only when the amount of membership overlap is significantly greater than the expected value. The presence of membership overlap means positive and/or negative interdependence between subunits. Positive interdependence occurs when membership experiences in multiple subunits are complementary or mutually reinforcing. Negative ecological interdependence exists because member time is a scarce resource for which organizational subunits compete in the high-choice environment. The allocation of time to one subunit potentially reduces the time available for participation in another subunit. In the context of ICA, interdependence between subunits is also influenced by situational factors, such as the creation of new interest groups or divisions, the policy for membership pricing (e.g., limit on the number of complimentary subunit memberships), and the evolution of research interests through member recruitment and attrition.

Concentration of membership overlap

One line of research in membership overlap assesses the level of concentration in networked inter-organizational ecologies, where competing and coexisting organizations (e.g., media outlets) are interconnected through the social affiliations of individuals (Fletcher & Nielsen, 2017; Mukerjee et al., 2018). Concentration is estimated using either an organization-centric or people-centric approach (Webster & Ksiazek, 2012). The former considers the distribution of market share, which is indicated by the Gini coefficient or other economic statistics (e.g., the Herfindahl–Hirschman index, Simpson’s diversity index). A high (or low) Gini coefficient is interpreted as a concentrated market where share percentage is unevenly (or evenly) distributed across organizations. As an application of this organization-centric approach in communication research, the theory of media niche (Dimmick, 2003; Webster, 2014) holds that the distribution of acquired resources (e.g., audience exposure, content revenue, and advertising revenue) across media organizations indicates the degree of niche differentiation and specialization in a media market.

In contrast, the people-centric approach takes interdependence into account and measures concentration by the extent to which the members of one organization overlap with those of others through network analysis. Although concentration and fragmentation are interrelated and capture different aspects of network structure, they are often viewed as opposite ends of the same spectrum (Webster & Ksiazek, 2012). Network concentration assesses the extent to which edges are unequally distributed among nodes. High concentration in the inter-organizational networks of membership overlap is characterized by a skewed distribution of the number of edges (i.e., connections representing membership overlap) for each node (i.e., organization). Structurally, high concentration represents a centralized configuration in which a few hub nodes have a disproportionately large number of edges, suggesting that certain organizations have membership overlap with many others. In contrast, low concentration is characterized by a similar number of edges across nodes (Webster & Ksiazek, 2012). The little variation in the number of edges suggests the absence of dominant organizations that share members with almost every other organization in the network.

Taking a people-centric approach, this study diverts attention from inter-organizational to intra-organizational membership overlap, with the goal of understanding the process of ecological

differentiation within organizations. In an intra-organizational network of membership overlap, nodes represent organizational subunits (e.g., ICA subunits), and structural concentration reflects the extent to which connections are unevenly distributed among subunits. A concentrated network structure indicates that a small number of hub subunits with a disproportionate share of edges are connected to many subunits, while connections between non-hub subunits are limited. Edges between subunits in the membership overlap network represent their high niche overlap after accounting for membership sizes. Consequently, the niches occupied by these hub subunits overlap considerably with those of many other subunits. In contrast, small niche overlap exists between non-hub nodes. Empirical support for network concentration suggests that hub subunits in the membership overlap network face competition or foster interdependence with a variety of generalists and/or specialists in the niche space.

The emergence of such hub nodes is explained by preferential attachment (Barabási & Albert, 1999), stating that nodes with more existing edges gain a competitive advantage in having even more edges in the future. This self-organizing feature suggests that an initial small difference in a subunit’s number of edges will be amplified over time, resulting in a concentrated structure where a small number of hub subunits become dominated in the edge distribution. In contrast, a lack of high concentration does not necessarily translate into network fragmentation (Borgatti, 2006); rather, it indicates that network edges will not be disproportionately attached to subunits that already have many edges. As a result, no single subunit will stand out with substantial membership overlap with many other subunits. The first research question asks whether concentration will be a defining feature in the evolution of the membership overlap network among subunits.

RQ₁: Will the formation of the membership overlap network among subunits be accompanied by network concentration?

Triadic closure of membership overlap

Network closure is the tendency to form a closed triad ($i-j$, $i-k$, $j-k$) among three nodes (i , j , k). Closure is often discussed together with brokerage (or lack of closure), which refers to the structure of an open triad ($i-j$, $i-k$) where i serves as an intermediary between otherwise disconnected others (Burt, 2005). Closure increases at the expense of brokerage within the same network. If a network consists of nodes whose neighbors are already connected, there will be little opportunity for brokerage.

Closure and brokerage provide different forms of network social capital (Burt, 2005; Leonardi & Contractor, 2018). If organizational subunits are represented as nodes, a structure rich in closed triads offers social and emotional support, fosters a sense of belonging and community, and enhances communication efficiency between subunits. In contrast, a brokerage structure enables certain subunits to act as bridges within the organization. This strategic position suggests a strong ability to access diverse information, generate creative ideas, and control the flow of resources and opportunities. Accordingly, members of brokerage subunits are more likely to integrate previously separated ideas and bring a new perspective on a familiar subject.

In an intra- or inter-organizational network of membership overlap, closure describes a situation where a set of three nodes (e.g., subunits or organizations) share a large percentage of members with each other. Despite the bridging benefits of brokerage positions, previous research has identified *triadic closure* as a self-organizing feature in the emergence of membership overlap networks. Even if a brokerage structure (represented by an open triad) exists in such networks, there will be a strong tendency for the open triad to be transformed into a closed triad at a later point in time. Consistent with the prevalence of closed triads predicted by triadic closure, Webster (2014) supported a theory of “massively overlapping culture” in which almost every media outlet shared a substantial portion of audience members with each other, offering limited opportunities for brokerage. Similarly, Xu (2022) found that triadic closure was an underlying mechanism for the occurrence of user overlap among social networking sites. Once a closed triad among the sites was created, it remained stable over time. Although membership overlap within an organization is different from that between organizations in many aspects (e.g., switching costs, participation barriers), boundary-maintaining mechanisms exist in both contexts to distinguish between members and non-members. In addition, network research has already shown that the mechanism of triadic closure applies to the self-organization of both intra- and inter-organizational networks (Monge & Contractor, 2003). Thus, it is hypothesized that two subunits that share the same third partner in the membership overlap network will be connected to each other.

H₁: The formation of the membership overlap network will be driven by triadic closure, leading to an increased likelihood of creating closed triads among three subunits.

Size dependence of membership overlap

Ecological research consistently supports the *liability of smallness* hypothesis, stating that smaller entities are more vulnerable to failure due to size disadvantages in resource acquisition (Baum & Amburgey, 2017). In the context of ICA, membership size is critical to subunit viability because a larger size means more budget and conference session slots allocated to a subunit. The liability of smallness is often discussed along with the *liability of newness*, which posits a higher risk of failure among younger entities (Stinchcombe, 1965). These two ecological liabilities can be intertwined because smaller entities tend to be younger in age. Nevertheless, evidence indicates that the observed age dependence disappears after controlling for size, while the size dependence predicted by the liability of smallness remains robust (Baum & Amburgey, 2017).

The liability-of-smallness theory suggests that stakeholders perceive small entity size as a negative indicator of the entity's legitimacy, past achievements, and future dependability (Baum & Amburgey, 2017). As a result, individuals associated with small entities lack the same level of influence and external endorsement as those affiliated with larger entities. If individuals aim to retain their existing social affiliations while mitigating the size disadvantages, a viable strategy is to seek multiple affiliations and avoid putting all eggs in one basket.

The higher chance of membership overlap for smaller entities is also explained by the phenomenon of *double jeopardy* (Ehrenberg et al., 1990; McPhee, 1963). Specifically, social

groups with a smaller market share or narrower appeal not only attract fewer new members but also engender less loyalty from them even after enrollment. Empirical research supporting double jeopardy often interprets aggregate group loyalty as the degree to which the group's members are also affiliated with other groups (Webster & Ksiazek, 2012). In an intra-organizational ecology where subunits are interconnected through membership overlap, a subunit's low level of loyalty is indicated by a high probability that its members are also members of another subunit. It is therefore predicted that:

H₂: A decrease in subunit size will lead to an increased likelihood of membership overlap with another subunit.

The liability-of-smallness theory also suggests that, once a small subunit experiences membership overlap, there is an increased likelihood that the subunit will share members with a larger subunit. For individuals who seek multiple affiliations without removing existing membership in a small subunit, joining a larger subunit will more effectively alleviate the potential disadvantages (e.g., lack of influence and external endorsement) they face. As size is interpreted as a legitimacy signal in organizational ecology (Baum & Amburgey, 2017), forging affiliations with a large subunit allows for borrowing legitimacy from the source. Another ecological explanation for the size dependence of membership overlap is that complementary differences (e.g., subunits with size differences) act as a force that pulls social entities together in the niche space (Baum & Haveman, 1997). Although H₂ predicts that large subunits are less likely to share members with another subunit, ecological complementarity suggests that once membership overlap occurs for a large subunit, its members tend to be affiliated with a smaller subunit. This is because membership in two subunits with significant size differences can better meet the needs for both uniformity and uniqueness. Therefore, there will be a higher chance of membership overlap between subunits with a larger difference in membership size.

H₃: Membership overlap will be more likely to occur between subunits dissimilar in size.

Network multiplexity and the interdependence of positive affinity relations and membership overlap

In social network research, *network multiplexity* refers to “the phenomenon in which more than one type of relationship exists between two nodes, with these multiple types of relationships potentially being interdependent, and influencing each other” (Lee & Lee, 2015, p. 57). In intra-organizational ecologies, network multiplexity means that the same set of organizational subunits are connected by multiple relations such as communication, collaboration, and partnership (Lee & Monge, 2011; Monge & Poole, 2008). From this multiplexity perspective, the formation of the membership overlap network within an organization is driven not only by personal choices for establishing multiple affiliations, but also by other relations between the same subunit pairs.

Affinity relations, defined as “socially constructed relationships that have either a positive or negative valence” (Shumate et al., 2013, p. 98), play an important role in intra-organizational ecologies. For instance, positive affinity

relations between subunits, such as alliances and partnerships, enable the sharing of scarce resources in the environment, contributing to the development of positive ecological interdependence (e.g., mutualism and symbiosis). In the research setting, positive affinity relations between ICA subunits are reflected in session co-sponsorship at an ICA annual conference. With limited session slots allocated for each ICA subunit, session co-sponsorship allows the sharing of session space, as well as other scarce resources (e.g., member time and attention) within the organization. This strategic alliance enhances a subunit's visibility to non-members without using its own session quota.

The mere exposure effect is the tendency for people to develop a preference for objects they are familiar with (Zajonc, 1968). Positive affinity relations between subunits make the mere exposure effect salient by raising the visibility of one subunit to members of another. The increased exposure enhances the knowledge and liking of non-members for the target, leading to a greater likelihood of dual membership. This positive association between affinity and membership overlap is also consistent with the concept of network multiplexity, suggesting that edge presence in one network (i.e., the membership overlap network) depends on edge presence in another network (i.e., the affinity network) among the same nodes (Lee & Lee, 2015).

H₄: There will be a positive association between the affinity network in year t and the membership overlap network in the same year.

Previous research on network multiplexity has called for a longitudinal design to investigate changes in the association between multiple networks created by the same set of nodes (Lee & Lee, 2015; Lee & Monge, 2011; Shumate et al., 2013). This study fills the gap by exploring the temporal dimension of network multiplexity in an intra-organizational ecology. The goal is to understand how the interdependence of the affinity network and the membership overlap network (predicted by H₄) will vary with time. Answering this question will provide insights into the evolution of membership overlap in response to other relations (e.g., positive affinity relations) in the same ecological space.

RQ₂: Will the strength of the positive association between the affinity network and the membership overlap network (H₄) change over time?

Methods

Data collection

Archival data on individual affiliations with subunits from 2007 to 2015 were obtained from ICA. The data were prepared as annual affiliation matrices, also known as two-mode network matrices (Wasserman & Faust, 1994), where rows represented ICA members and columns represented subunits. Binary cell values in the nine annual matrices were coded as 1 if ICA members were affiliated with a specific division or interest group in year t , and 0 otherwise.

The list of individual members experienced year-to-year fluctuations as new members joined, and existing members left ICA and its subunits. Each individual member had a unique ID. Personal identification and other sensitive

information, such as name, date of birth, gender, home and email addresses, country of residence, and institutional affiliation, were unavailable for research purposes.

The list of ICA subunits was fixed. The analysis included a total of 24 ICA divisions and interest groups founded no later than 2007. The four interest groups (Environmental Communication; Media Industry; Mobile Communication; Sports Communication) founded from 2008 to 2015 were excluded due to the assumption of a fixed set of nodes available for connections at all time points in temporal models for organizational networks (Leifeld et al., 2018).

Supplementary Table S1 presents the membership sizes of each ICA subunit and the entire association for the years 2007 to 2015. All individuals with membership status in any year of the study period were included in the analysis. The "total number of ICA members" row excludes those without any subunit affiliations. The average number of affiliated subunits (excluding affiliations with the four interest groups founded from 2008 to 2015) at the individual level ranged from 2.54 (the year 2007) to 2.75 (the year 2010). The proportion of ICA members with 1–6 subunit affiliations varied between 95% and 96%. The Gini coefficient (0 = complete equality, 1 = complete inequality) was computed to assess the level of equality in the distribution of subunit membership sizes. The Gini coefficient ranged from .29 to .31 and gradually decreased over time (see **Supplementary Table S1**), showing little evidence of the concentration of membership sizes and its intensification within ICA.

Processing of membership data

The annual affiliation matrices were transformed into symmetric 24×24 matrices, where both rows and columns represented the ICA divisions and interest groups. The value of a diagonal cell (i, i) was the number of ICA members affiliated with subunit i in year t . In contrast, the value of an off-diagonal cell (i, j) was the number of individuals who were members of both i and j in the same year.

The outcome of interest in the temporal network model is a binary network among the same nodes, observed repeatedly at multiple time points. To construct the dependent variable for the study, the nine annual 24×24 network matrices obtained from ICA were transformed into a binary network dataset, where edges were defined based on a threshold of connection strength (Mukerjee et al., 2018). As a self-loop indicates membership overlap between a subunit and itself, which is considered meaningless, all diagonal cell values were set to zero. The procedures for binary value assignment in network construction and for the sensitive check are described in **Supplementary Materials**.

Table 1 presents the descriptive statistics of the longitudinal network of membership overlap after the threshold was applied. Despite the relative stability of the total number of ICA members over time (ranging from 3,687 to 4,271), the overall trend is a noticeable increase in network density and transitivity, and a noticeable decrease in the average degree of separation and modularity. **Supplementary Table S2** reports an ICA subunit's number of edges (a.k.a. degree centrality) in the membership overlap network for each year from 2007 to 2015 after the threshold was applied. **Supplementary Table S3** reports the betweenness centrality of each ICA subunit over the study period. **Figure 2** depicts a network visualization of the 9-year aggregate structure of membership overlap.

Measures

Dependent variable

Membership overlap in year t was represented by binary cell values in the network matrices generated in the previous section using the threshold approach. A cell value of 1 (edge presence) indicated that the observed membership overlap between ICA subunits i and j in year t was significantly higher than the expected value (Mukerjee et al., 2018). The column of network density in Table 1 showed that, after the filtering process, present edges accounted for 47–67% of the possible pairwise combinations of the 24 ICA subunits over the study period.

Independent variables

Network concentration (RQ₁) was represented by the term of geometrically weighted degree (gwdegree). A term denotes a specific network feature and is included as a predictor in network models that account for the interdependence among nodes.

The gwdegree term models the tendency of edges to be dispersed across nodes over time. A negative and significant estimate of the gwdegree term indicates an increased likelihood of edges being disproportionately attached to nodes with a higher number of existing edges, resulting in a more concentrated network structure.

Triadic closure (H₁) was indicated by the triangle term, which reflects the emergence of closed triads in an undirected network through the mechanism of triadic closure. As previously noted, a closed triad describes a set of three nodes (i, j, k) in which all nodes have edges with each other ($i-j, i-k, j-k$). A positive and significant estimate of the triangle term provides evidence for the tendency of three nodes to form a closed triad in the observed network over time.

Subunit size (H₂ and H₃) was measured by the number of ICA members affiliated with the given division or interest group in year t . As reported in Supplementary Table S1, the variable ranged from 50 to 1,412, with an average of 444.13 and a standard deviation of 265.24. The Mass Communication Division had the largest membership size throughout the 9 years.

Positive affinity relations between subunits (H₄ and RQ₂) were determined by whether i and j jointly sponsored a

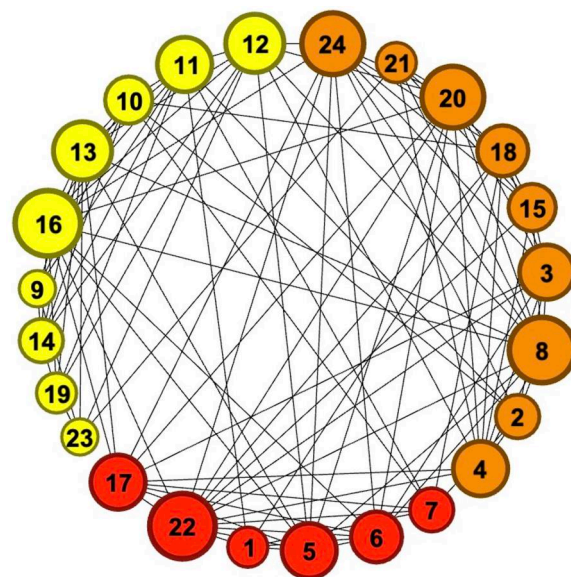


Figure 2. A network visualization of the structure of membership overlap among ICA subunits.

Note. The graph was created using Gephi version 0.10.1 with the Circular layout. For improved readability, the nine binary matrices generated by the filtering algorithm were combined into a single network. In this network, an undirected edge was added to the graph only when the observed membership overlap between two ICA subunits was significantly greater than the expected value across all 9 years of the study. As a result, the presence of only a limited number of individuals with active membership in two ICA subunits throughout the 9-year period was insufficient to create a connection between the nodes in the graph. Node sizes are proportional to degree centrality. Node labels correspond to the assigned subunit numbers in Supplementary Tables S1, S2, and S3 (1 = Children, Adolescents, and the Media; 2 = Communication and Technology; 3 = Communication History; 4 = Communication Law and Policy; 5 = Ethnicity and Race in Communication; 6 = Feminist Scholarship; 7 = Game Studies; 8 = Global Communication and Social Change; 9 = Health Communication; 10 = Information Systems; 11 = Instructional & Developmental Communication; 12 = Intercultural Communication; 13 = Intergroup Communication; 14 = Interpersonal Communication; 15 = Journalism Studies; 16 = Language and Social Interaction; 17 = Lesbian, Gay, Bisexual, Transgender & Queer Studies; 18 = Mass Communication; 19 = Organizational Communication; 20 = Philosophy, Theory and Critique; 21 = Political Communication; 22 = Popular Communication; 23 = Public Relations; 24 = Visual Communication Studies). Node colors indicate the modularity classes assigned by Gephi's community detection algorithm, which partitions the network into clusters of densely connected nodes (Blondel et al., 2008). Three clusters were identified, with a modularity score of .25. Nodes with the same color are more interconnected with each other than with nodes in different colors.

Table 1. Descriptive statistics of the observed network of membership overlap, 2007 to 2015

Time point	Number of present edges	Network density ^a	Number of closed triads ^b	Transitivity ^c	Average degree of separation ^d	Modularity ^e
2007	131	.47	281	.60	1.55	.21
2008	160	.58	484	.69	1.43	.14
2009	149	.54	389	.64	1.47	.15
2010	155	.56	441	.66	1.45	.15
2011	147	.53	400	.66	1.48	.14
2012	163	.59	510	.69	1.41	.12
2013	166	.60	540	.70	1.40	.13
2014	178	.64	635	.73	1.36	.09
2015	185	.67	692	.74	1.33	.08

Note.

^a Network density is the ratio of the number of present edges to the number of possible edges. An undirected edge is present if the observed membership overlap between two nodes is significantly greater than the threshold value set by the filtering algorithm. The number of possible edges at each time point is calculated as 276 , using the formula $n \times (n-1)/2$ in which $n = 24$.

^b A closed triad is a set of three nodes in which all nodes are connected to each other.

^c Transitivity ranges from 0 to 1. A higher transitivity score indicates a greater likelihood that two nodes connected to the same third node are directly connected.

^d Average degree of separation (a.k.a. average shortest path length) is the average number of steps required for a node to reach another in a network. In a fully connected network where every node is directly connected to one another, the average degree of separation is 1.

^e Modularity ranges from 0 to 1. A higher modularity score indicates a better division of the network into separate subcommunities.

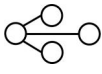
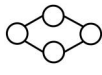
session at the ICA conference in year t . Based on archival data on session co-sponsorship retrieved from online conference programs on the ICA site, the variable was represented by a binary network matrix in year t to predict edge presence in the membership overlap network in the same year. Cell values were coded as 1 if two subunits co-sponsored at least one session (e.g., a session of regular or interactive papers, or a panel or reception session) at an annual conference. Values were coded as 0 if a pair of subunits did not co-sponsor any session in year t . The density of the annual network matrices ranged from .07 (the year 2010) to .13 (the year 2007), with an average of .11 and a standard deviation of .02.

Control variables

Time was included to capture temporal variations in the presence of edges in the membership overlap network. In addition, four structural configurations and two subunit attributes were controlled in the analysis. The edges term is the baseline tendency of edge presence in a longitudinal network. The 2-stars term represents an open triad involving three nodes (i, j, k) with edges present between i and the other two nodes ($i-j, i-k$). The term is non-discriminatory because it can be interpreted as either network centralization or brokerage within an undirected triad. It must be controlled to serve as a baseline estimate for a centralized or brokerage structure at the local level. The 3-stars term represents a set of four nodes (i, j, k, l) in which i is linked to j, k , and l ($i-j, i-k, i-l$), suggesting a tendency of network centralization among four nodes. As an alternative network structure among the same number of nodes, the 4-cycle term describes a configuration in which i and j do not have a direct edge but are indirectly linked through k and l ($i-k, i-l, j-k, j-l$). A negative and significant estimate of the 4-cycle term reflects the lack of network brokerage among four nodes, providing additional evidence for the presence of network closure. The graphical representations of the edges, 2-stars, 3-stars, and 4-cycle terms are displayed in Table 2.

As for subunit attributes, *subunit age* was measured by the difference between the year of observation and the year when the division or interest group was founded in ICA. The founding time ranged from the year 1968 (Information Systems; Interpersonal Communication; Mass Communication; Organizational Communication) to the year 2007 (Children, Adolescents, and the Media; Communication History). Considering the possibility of the liability of newness (Stinchcombe, 1965), this variable controlled for the age dependence of membership overlap. *Subunit status* was a time-varying binary variable, indicating whether a subunit was a division (coded as 1) or interest group (coded as 0) in year t . An interest group represents an emerging scholarly interest and has a membership rate lower than 5% (equivalent to about 200 members) in ICA, while a division often evolves from an interest group. ICA divisions tend to have larger membership sizes and are older in age than interest groups, thus covering some aspects of subunit size and newness. Five subunits (Children, Adolescents, and the Media; Communication History; Ethnicity and Race in Communication; Game Studies; Journalism Studies) changed their status from interest groups to divisions during the 9-year period. The other subunits (e.g., the Communication and Technology Division, the Intergroup Communication Interest Group) had the same status throughout the study period.

Table 2. Results of the TERGM predicting membership overlap

	Model 1 (baseline)		Model 2 (full)	
	Estimate	SE	Estimate	SE
Terms for structural configurations				
Edges	1.297***	0.211	1.387	1.475
				
2-stars			-0.233*	0.103
				
3-stars			0.010	0.007
				
4-cycle			-0.047***	0.006
				
RQ ₁ : Gwdegree			1.755	1.088
H ₁ : Triangle			1.056***	0.053
Terms for node attributes				
Subunit age (nodecov)	-0.017***	0.003	-0.002	0.002
Subunit age (absdiff)	-0.026***	0.004	-0.013***	0.003
Subunit status (nodecov)	-0.333	0.183	-0.165	0.157
Subunit status (nodematch)	0.420*	0.195	0.429*	0.173
H ₂ : Subunit size/100 (nodecov)			-0.107***	0.020
H ₃ : Subunit size/100 (absdiff)			0.044*	0.022
Terms for edge attributes				
Time (timecov)	0.117***	0.017	0.128***	0.031
H ₄ : Affinity (edgescov)			1.114***	0.316
RQ ₂ : Affinity × Time (timecov)			0.022	0.061
Area under the precision-recall curve	.675		.763	

Note.

* $p < .05$, *** $p < .001$.

Analytical procedures

Temporal exponential random graph models (TERGMs), which extend traditional ERGMs for discrete-time analysis, were employed to test all hypotheses (Leifeld et al., 2018). The analysis was performed using the *btergm* package (version 1.10.10) in R (version 4.3.1). The package uses maximum pseudolikelihood estimation (MPLE) with bootstrapped confidence intervals or Markov chain Monte Carlo maximum likelihood estimation (MCMC-MLE) to make statistical inferences. While MCMC-MLE is more computationally demanding than MPLE, it offers more accurate estimates and produces better model fit, especially when the number of time steps is small. As a result, TERGMs were run by MCMC-MLE through the *mtergm* function in the *btergm* package. Another longitudinal model for

network panel data within the ERGM framework is separable temporal ERGMs (STERGMs; Krivitsky & Handcock, 2014). However, STERGMs cannot model the probability of edge presence as a function of an edge attribute and its interaction with time, making it impossible to test H_4 and answer RQ_2 .

All independent and control variables were entered as network terms for *structural configurations*, *node attributes*, or *edge attributes* in the TERGM analysis. Terms for structural configurations account for complex dependencies among edges, emphasizing that network structures emerge from inherent self-organizing processes. The structural configurations included in the analysis were gwdegree, triangle, edges, 2-stars, 3-stars, and 4-cycle.

Terms for node attributes in undirected networks include nodecov, absdiff (an abbreviation for absolute difference), and nodematch. A nodecov term determines whether a subunit with a higher value for a given attribute is more likely to form an edge with another subunit. Absdiff and nodematch terms are used for numerical (i.e., subunit size and age) and categorical (i.e., subunit status) node attributes, respectively. A negative and significant estimate of an absdiff term (representing similarity in a numerical attribute) and a positive and significant estimate of a nodematch term (representing similarity in a categorical attribute) indicate that subunits are more likely to be linked to those with similar or identical characteristics.

Terms for edge attributes include edgecov and timecov. An edgecov term determines whether edge presence in the observed network is predicted by edge presence in another network among the same nodes. A timecov term estimates (a) linear time trends in the probability of edge presence, or (b) the interaction effect between time and another network. The edge attributes in the TERGM analysis were the affinity network among the 24 ICA subunits, time, and their interaction. Table 3 provides a summary of TERGM interpretations for research questions and hypotheses.

Results

Table 2 presents the results of the TERGMs predicting membership overlap in the longitudinal network observed at nine discrete time points. The MCMC sample size and MCMC interval were set as 5,000 and 2,000, respectively. For ease of interpretation, the value of subunit size was divided by 100 prior to analysis. A network term is statistically significant if the ratio of its coefficient to standard error exceeds 1.96 in magnitude, which corresponds to a 95% confidence interval.

Estimates in both TERGMs and traditional logistic regression represent the log-odds of an outcome and reflect the magnitude of an effect. However, unlike the interpretation of a coefficient estimate as a change in the log-odds for a one-unit change in the predictor in logistic regression, a term estimate for a structural configuration represents the log-odds of its occurrence in the observed network compared to that in the simulated networks. For a term for a node or edge attribute, its estimate in TERGMs is interpreted as a change in the log-odds of edge formation in the observed network associated with a one-unit change in the attribute.

Without accounting for structural dependencies, Model 1 provides baseline estimates for the controlled node and edge attributes. Model 2 adds independent variables and alternative mechanisms for network self-organization to the baseline model. The results of another potential baseline model with all controlled terms included as predictors were not reported due to the non-identifiability of term estimates. A consistent pattern from Models 1 and 2 was the increased probability of edge presence over the nine years, as evidenced by the positive and significant estimate of the time effect. Age and status homophily were also supported by (a) the negative and significant absdiff term for subunit age and (b) the positive and significant nodematch term for subunit status. As a result, membership overlap tended to exist between subunits with similar age or with the same status.

The term estimates from Model 2 were used for hypothesis testing. As a validity check for the TERGM estimation, the

Table 3. TERGM interpretations for research questions and hypotheses

Theoretical mechanism	Structural and attribute predictors of membership overlap	The corresponding TERGM term	Interpretation of a supportive result for ICA
Network concentration of membership overlap (RQ_1)	Dispersion of edges across subunits in the membership overlap network	The gwdegree term	A negative and significant estimate indicates that ICA subunits with more existing edges in the membership overlap network are disproportionately attached to more edges over time.
Triadic closure of membership overlap (H_1)	Probability of forming closed triads in the membership overlap network	The triangle term	A positive and significant estimate indicates that two ICA subunits that share the same partner in the membership overlap network are more likely to be connected to each other over time.
Size dependence of membership overlap (H_2)	Size of a single subunit	The nodecov term for subunit size	A negative and significant estimate indicates that smaller ICA subunits have more edges in the membership overlap network.
Size dependence of membership overlap (H_3)	Size difference between two subunits	The absdiff term for subunit size	A positive and significant estimate indicates that edges in the membership overlap network are more likely to be formed between two ICA subunits with dissimilar membership sizes.
Network multiplexity (H_4)	Presence of an edge between two subunits in the affinity network	The edgecov term for affinity	A positive and significant estimate indicates that two ICA subunits are more likely to be connected in the membership overlap network if they are already connected in the affinity network.
Network multiplexity (RQ_2)	Time-varying effect of edge presence in the affinity network	The timecov term for affinity $\times$ time	A positive and significant estimate indicates that the association between the affinity network and the membership overlap network is stronger over time.

goodness-of-fit (GoF) of Model 2 was evaluated using the area under the curve (AUC) value (Leifeld et al., 2018). In Table 2, the AUC-precision recall value of the full model (.763) was 13% higher than that of the baseline model (.675), providing evidence of reasonably accurate predictions. The GoF assessment also involved comparing higher-order statistics (e.g., dyad-wise shared partners, edge-wise shared partners, degree, and geodesic distances) of the observed network with those of the simulated networks (Leifeld et al., 2018). Supplementary Figure S1 displays the plots of GoF statistics. Solid lines in the graph represent actual values, while boxplots represent the distribution of simulated values. The GoF test showed that more than 90% of the observed values were not statistically different from the values simulated by the algorithm at the .05 level, indicating a good model fit.

RQ₁ asked whether the formation of the membership overlap network among subunits would be accompanied by network concentration. The results showed that the estimate of the gwdegree term was not statistically significant (estimate = 1.755, SE = 1.088, $p = .11$). Edges were not concentrated on a small number of subunits in network evolution.

H₁ predicted that the formation of the membership overlap network would be driven by triadic closure, leading to an increased likelihood of creating closed triads among three subunits. The results revealed that the estimate of the triangle term was positive and significant (estimate = 1.056, SE = 0.053, $p < .001$), meaning that the closure configuration occurred more frequently in the observed network than in the simulated networks generated by the estimation algorithm. This result provided support for H₁.

H₂ posited that a decrease in subunit size would lead to an increased likelihood of membership overlap with another subunit. As predicted, the estimate of the nodecov term for subunit size was negative and significant, lending support to H₂ (estimate = -0.107, SE = 0.020, $p < .001$). A one-unit increase in a subunit's membership size (equivalent to 100 members) decreased the odds of membership overlap with another subunit by 10% ($1 - \exp(-0.107)$).

H₃ stated that membership overlap would be more likely to occur between subunits dissimilar in size. The hypothesis would be supported if the estimate of the absdiff term for subunit size was positive and significant. The results were consistent with the prediction that a larger absolute difference in size would be associated with an increased probability of edge formation in the observed network (estimate = 0.044, SE = 0.022, $p = .049$).

H₄ predicted a positive association between the affinity network in year t and the membership overlap network in the same year. The results demonstrated that the estimate of the edgcov term was positive and significant (estimate = 1.114, SE = 0.316, $p < .001$), supporting H₄. The presence of an edge in the affinity network would increase the odds of edge presence in the membership overlap network by 205% ($\exp(1.114) - 1$).

RQ₂ asked whether the strength of the positive association between the affinity network and the membership overlap network would change over time. The results showed that the timecov term for Affinity $\times$ Time was not statistically significant (estimate = 0.022, SE = 0.061, $p = .36$). There was no evidence that the correlation between the two network matrices varied linearly with time.

Discussion

The objective of this study is to explore the formation mechanisms of membership overlap in an intra-organizational ecology, where competing and coexisting subunits are interconnected through shared members. Consistent with the conceptualization of ecological interdependence as network ties (Monge & Poole, 2008) and the treatment of overlap data as network data (Mukerjee et al., 2018), this study understands changing niche overlap among the 24 ICA subunits as a longitudinal network of membership overlap. Edge presence in the network indicates a substantial degree of membership overlap between two subunits in a given year.

Variations in membership overlap over time, which are reflected in edge changes in the membership overlap network, are planned or blind (Aldrich et al., 2020). Planned variations arise from deliberate human intent (e.g., membership choices guided by personal interests, strategic partnerships between subunits), while blind variations occur randomly and are beyond the control of human agency. The TERGM framework (Leifeld et al., 2018) provides a statistical tool for controlling network change resulting from blind variations. It does so by assessing the extent to which the structure of the observed network departs from that of the simulated networks. The results of the TERGM analysis show that the creation of membership overlap in the intra-organizational ecology of ICA is driven by subunit demographics (i.e., size dependence) and structural interdependence within the same network (i.e., triadic closure) and between different networks (i.e., network multiplexity).

Structure of the membership overlap network: closure without concentration

Webster and Ksiazek (2012) proposed that the concentration of individual choices of social objects (e.g., organizational subunits) could be evaluated through (a) the popularity distribution of objects (e.g., the concentration of subunit membership sizes) and (b) the degree of affiliation overlap among objects (e.g., the concentration of the membership overlap network between subunits). In the context of ICA, the former is indicated by the level of inequality in the distribution of subunit membership sizes. In contrast to a winner-take-all pattern, the consistently low Gini coefficient and its stability over time (see Supplementary Table S1) provide little evidence of the concentration of membership sizes within ICA.

Regarding the degree of affiliation overlap, the non-significant gwdegree term (see Table 2) shows no tendency toward network concentration in the evolution of the membership overlap network among ICA subunits (RQ₁). Consequently, no single ICA division or interest group attracts a disproportionately large number of connections over time. An ecological interpretation of the finding is that the structure of membership overlap is not dominated by a small number of hub subunits whose occupied niches in the intra-organizational ecology overlap considerably with those of many other ICA subunits. If each ICA subunit is considered as a subfield of communication studies (Doerfel & Barnett, 1999), failure to support high concentration in the membership overlap network reflects the lack of a core sub-area broad enough to integrate different specializations or niche topics.

Despite the lack of support for network concentration, the membership overlap network demonstrates a self-organizing

feature of triadic closure, as evidenced by the higher prevalence of closed triads in the observed network than in the simulated networks generated by the TERGM (H_1). This finding suggests that the internal structure of ICA is becoming cohesive rather than siloed over time. A contextual factor potentially contributing to the formation of closed triads through the closure mechanism is that ICA allowed registered members to join up to three divisions or interests at no extra cost. It was speculated that this incentive led to an increase in the number of subunit pairs with a connection strength greater than the threshold value set by the filtering algorithm. However, not all ICA members took advantage of the availability of free memberships in three subunits. The descriptive results from the last two rows of [Supplementary Table S1](#) indicate that the average number of affiliated subunits for an ICA member is below 3, ranging from 2.54 to 2.75.

A reasonable explanation for the increased density and transitivity of the network (see [Table 1](#)), and the few occurrences of the 2-stars (representing an open triad) and 4-cycle (representing a closed loop composed of two open triads) structures (see [Table 2](#)) is that edges added to the membership overlap network mostly filled the available open triads through triadic closure, increasing the number of closed triads over time. For instance, Communication and Technology shared members with the two disconnected subunits (e.g., Communication History and Instructional & Developmental Communication) in the membership overlap network in 2010. Triadic closure facilitated the sharing of members between Communication History and Instructional & Developmental Communication in later years, leading to the formation of a closed triad among the three ICA subunits. The conversion of open triads into closed triads is often interpreted as increased closure at the expense of brokerage ([Burt, 2005](#)). Although structural cohesion among ICA subunits promotes the development of collective identity and solidarity, reduced brokerage limits opportunities for bridging across subunits and generating innovative ideas through knowledge integration ([Leonardi & Contractor, 2018](#)). A certain degree of brokerage within ICA may be essential for driving growth and innovation in the communication field, as high-impact research often involves atypical combinations of existing knowledge, providing a cross-boundary perspective on a familiar subject ([Uzzi et al., 2013](#)). The ideal balance between closure and brokerage among ICA subunits is open for further debate.

Subunit size and positive affinity as attribute features of the membership overlap network

The results concerning the effects of node and edge attributes on edge formation show that (a) the sizes of individual subunits and subunit pairs and (b) affinity relations between subunits drive the creation of membership overlap. In accordance with the liability of smallness ([Baum & Amburgey, 2017](#)) and double jeopardy ([Ehrenberg et al., 1990](#); [McPhee, 1963](#)), this study provides empirical support for the size dependence of membership overlap. Specifically, subunits with fewer members are more likely to share a significant portion of their members with other subunits, as evidenced by the negative and significant nodecov term for subunit size (H_2). This negative association between subunit size and membership overlap appears after accounting for the age dependence predicted by the liability of newness ([Stinchcombe, 1965](#)). In contrast, the negative nodecov term for subunit age observed in the baseline model

(estimate = -0.017 , $SE = 0.003$, $p < .001$) loses significance in the full model (estimate = -0.002 , $SE = 0.002$, $p = .32$; see [Table 2](#)), confirming that the age dependence of membership overlap is spurious. Subunit status is another control variable associated with subunit size. The non-significant nodecov term for subunit status in the full model shows that subunit size is a more robust legitimacy predictor of membership overlap than subunit status.

Another finding of size dependence is that membership overlap is more prevalent between subunits with a larger difference in membership size (H_3). This sheds light on the size characteristics of target subunits to which a smaller subunit is attached when it has a higher probability of sharing members with another subunit (H_2). The evidence from the positive absdiff term (H_3) and the negative nodecov term (H_2) in Model 2 supports both hypotheses, indicating that small-sized subunits have a stronger dependence on large-sized ones than vice versa within ICA. A small membership size signals a lack of legitimacy, dependability, and mass appeal ([Baum & Amburgey, 2017](#)). Affiliations with both small- and large-sized subunits can be complementary, allowing members of small subunits to compensate for the associated size disadvantages (e.g., the subunit's lack of influence and external endorsement). If a subunit's number of edges in the network is interpreted as the aggregate level of member loyalty to the subunit ([Webster & Ksiazek, 2012](#)), the observed size dependence suggests that small ICA subunits rarely attract focused and exclusive members.

In response to the call for a longitudinal examination of network multiplexity ([Lee & Lee, 2015](#); [Lee & Monge, 2011](#); [Shumate et al., 2013](#)), this study provides evidence of the interdependence of the affinity network and the membership overlap network. Specifically, membership overlap is more likely to occur between ICA subunits that co-sponsor at least one session at an ICA annual conference (H_4), suggesting that positive affinity relations contribute to membership overlap through increased familiarity and liking for the target subunit. Although the results of the post-hoc analysis in [Supplementary Table S4](#) do not completely preclude the possibility of reverse causality from membership overlap to affinity, the path from affinity to membership overlap became dominant in the 2012 to 2015 period. This shift may be a result of broader initiatives by ICA.

Another important finding is the stable positive association between the affinity network in year t and the membership network in the same year, showing no significant fluctuations over time (RQ_2). The observed stability offers a more nuanced understanding of how membership overlap responds and adapts to other relations (e.g., positive affinity relations) in the same intra-organizational ecosystem. In the evolution of the membership overlap network, the absence of a time-varying effect indicates that there is no selective force in favor of or against edge presence in the affinity network. A practical implication for ICA is that, while the decision to hold membership in two ICA subunits is guided by the presence of session co-sponsorship between them, this community preference does not grow stronger or weaker over time.

Implications

This study leverages the ecological approach to group and organizational communication ([Lee & Monge, 2011](#); [Monge & Poole, 2008](#)) by interpreting niche overlap among organizational subunits as an intra-organizational network of

membership overlap. The incorporation of a network perspective into research on ecologies of social affiliations (McPherson, 2004; McPherson et al., 2001) provides a relational account of longitudinal changes in niche overlap among social entities. In addition, the network investigation of the formation mechanisms of membership overlap within ICA contributes to developing a middle-range niche theory that connects abstract ecological terms with concrete intra-organizational phenomena, advancing the understanding of the structure and evolution of ICA and the communication field. Moreover, the positive association between affinity and membership overlap demonstrates the potential for communication research on membership and audience overlap (Mukerjee et al., 2018; Webster, 2014; Webster & Ksiazek, 2012) to gain valuable insights from the perspective of network multiplexity. This multiplexity approach emphasizes that structural interdependence exists not only within the same network but also between different networks among the same nodes (Lee & Lee, 2015; Shumate et al., 2013).

In addition, this study expands the understanding of the temporal network structure of ICA and the communication field (Barnett & Danowski, 1992; Barnett & Park, 2023; Chung et al., 2009; Doerfel & Barnett, 1999). Speaking across communication subfields is achieved through new combinations of existing elements (Tenenboim-Weinblatt & Lee, 2020), suggesting that knowledge integration involves converting network brokerage to closure. Bibliometric analyses often use (a) structural equivalence in citation networks (Doreian & Fararo, 1985) or (b) co-citation (Borgman & Rice, 1992) as an indicator of niche overlap between academic journals, offering insights into unconnected sub-literatures that need to be bridged. This study adds to the closure and brokerage structures of niche overlap in the communication field by identifying the patterns of membership overlap among ICA subunits. Fragmentation is characterized by the coexistence of high closure within clusters and high brokerage between clusters (Song et al., 2020). In the context of ICA, a cluster consists of divisions and/or interest groups that bond together through membership overlap or other relationships. If the internal structure of ICA is fragmented, there is a separation between multiple internally cohesive clusters, suggesting the presence of sufficient bridging opportunities. The identification of closure as a self-organizing feature in the formation of membership overlap, along with a low average degree of separation and modularity score (see Table 1), does not support the notion that the communication field has been increasingly fragmented. Instead, the findings on the internal structure of ICA are consistent with Song et al.'s (2020) observation of "a surprisingly encouraging degree of cohesion and reachability across different subfields of communication science research" (p. 327) from a simulation analysis of interconnections among research topics.

While the high closure limits opportunities for brokerage within ICA, the analytical approach provides guidance on how to identify subunit pairs that have yet to be bridged. Supplementary Table S5 lists the pairs of ICA subunits with observed overlap significantly smaller than the expected value. Statistical significance was determined by whether the corresponding t_{ij} value was less than -2.58 ($p < .01$) using the same threshold approach described in the methods section. Many disconnected pairs in Supplementary Table S5 reflect the historical separation between mass-oriented (e.g., Journalism Studies, Mass Communication, and Political Communication) and

interpersonal-oriented (e.g., Health Communication and Interpersonal Communication) subunits within ICA (Barnett & Danowski, 1992). The evidence for network multiplexity in the creation of membership overlap suggests that ICA leadership teams can strategically build affinity relations between the disconnected subunits to (a) increase between-group exposure to heterogeneous information and (b) promote new collaboration and integration of previously separated areas. Although the recommendations are intended to reduce network brokerage by bridging unconnected ICA subunits, the lack of connection and its persistence are advantageous for individuals exploiting information asymmetries between subunits for transferring knowledge and identifying research gaps and new opportunities.

Limitations and directions for future research

TERGMs are limited to analyzing only binary network data as dependent variables (Leifeld et al., 2018). Although the original network data on membership overlap were weighted, cell values in the network matrices were dichotomized before running the TERGM. Exponential-family models for valued networks are currently less developed than those for binary networks due to the computational cost of estimation. When the weighted version of TERGMs is available, researchers should check the robustness of the results by considering edge strength in network modeling.

The focus on the intra-organizational ecology of ICA means that the findings are not necessarily generalizable. While ICA is a professional and non-governmental association, its internal structure where multiple subunits coexist and compete for scarce resources (e.g., the limited time and effort of members) is representative of the current organizational landscape. Like ICA, many traditional organizations and online communities allow individuals to join and leave subunits (e.g., departments, teams, and groups) at low cost, leading to substantial membership overlap within organizational boundaries (Wang et al., 2013). Future research should determine whether the observed patterns of membership overlap exist in other organizations or organizational forms.

The historical data on membership overlap obtained from ICA before the COVID-19 pandemic do not contain information about individual affiliations with subunits in recent years. The pandemic disrupted and complicated the follow-up data collection, resulting in the study period ending in 2015. It is noteworthy that the number of divisions and interest groups within ICA has increased from 28 (24 included in the analysis, plus 4 founded during the observation period) to 33 since the year 2015. The process of organizational speciation (Weber, 2017) suggests that the emergence of a new division or interest group disrupts the stability of the intra-organizational ecosystem. An important area for future research is how membership overlap between ICA subunits adapts and responds to such disruptions. Although extending the study period to include more recent years could help smooth year-to-year fluctuations, TERGMs operate on the assumption of a fixed node set (Leifeld et al., 2018). This means that the set of ICA subunits included in the analysis must have existed at the start of the study period and remained available for connections in each subsequent year. Even with an extension of the study period beyond 2015, it would still be impractical to assess the impacts of newly founded ICA subunits on the internal structure of ICA in a single TERGM analysis.

Another limitation of the ICA data is the absence of socio-demographic information at the individual level, making it impossible to place each subunit in the multidimensional niche space that describes the distribution of social resources (McPherson, 1983). The ecological model of affiliation (McPherson, 2004) predicts that a person's position relative to an entity's niche affects the likelihood of joining or maintaining membership with the entity. Future research on membership overlap should use individual socio-demographic data to determine whether potential and existing members located at the center or periphery of a subunit's niche are more likely to be recruited and retained.

Conclusion

This article elucidates the mechanisms underlying the formation of membership overlap in the intra-organizational ecology of ICA. The results indicate that no single division or interest group stands out with extensive membership overlap with many other ICA subunits. Triadic closure facilitates the creation of a cohesive structure among subunits that limits opportunities for brokerage. In addition, an ICA subunit with a smaller membership size has a greater likelihood of sharing members with another, particularly with a larger subunit, suggesting an asymmetric dependence of small-sized subunits on large-sized ones. Furthermore, the existence of an affinity relation between two ICA subunits is positively associated with membership overlap between them. However, the return value of this affinity relationship (represented by session co-sponsorship at an ICA conference) does not change over time.

Supplementary material

Supplementary material is available online at *Journal of Communication* online.

Data availability

The data underlying this article are owned by ICA. Data will be shared upon request to the author, subject to ICA's permission.

Conflicts of interest: None declared.

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References

- Aldrich, H. E., Ruef, M., & Lippmann, S. (2020). *Organizations evolving* (3rd ed.). Edward Elgar Publishing.
- Barabási, A. L., & Albert, R. (1999). Emergence of scaling in random networks. *Science*, 286(5439), 509–512. <https://doi.org/10.1126/science.286.5439.509>
- Barnett, G. A., & Danowski, J. A. (1992). The structure of communication: A network analysis of the International Communication Association. *Human Communication Research*, 19(2), 264–285. <https://doi.org/10.1111/j.1468-2958.1992.tb00302.x>
- Barnett, G. A., & Park, H. W. (2023). Co-authorship among the fellows of the International Communication Association. *Scientometrics*, 128(6), 3401–3418. <https://doi.org/10.1007/s11192-023-04705-6>
- Baum, J. A. C., & Amburgey, T. L. (2017). Organizational ecology. In J. A. C. Baum (Ed.), *The Blackwell companion to organizations* (pp. 304–326). John Wiley & Sons, Ltd <https://doi.org/10.1002/9781405164061.ch13>
- Baum, J. A. C., & Haveman, H. A. (1997). Love thy neighbor? Differentiation and agglomeration in the Manhattan hotel industry, 1898–1990. *Administrative Science Quarterly*, 42(2), 304–338. <https://doi.org/10.2307/2393922>
- Blondel, V. D., Guillaume, J. L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10), P10008. <https://doi.org/10.1088/1742-5468/2008/10/P10008>
- Borgatti, S. P. (2006). Identifying sets of key players in a social network. *Computational and Mathematical Organization Theory*, 12(1), 21–34. <https://doi.org/10.1007/s10588-006-7084-x>
- Borgman, C. L., & Rice, R. E. (1992). The convergence of information science and communication: A bibliometric analysis. *Journal of the American Society for Information Science*, 43(6), 397–411. [https://doi.org/10.1002/\(SICI\)1097-4571\(199207\)43:6<397::AID-ASII>3.0.CO;2-M](https://doi.org/10.1002/(SICI)1097-4571(199207)43:6<397::AID-ASII>3.0.CO;2-M)
- Burt, R. S. (2005). *Brokerage and closure: An introduction to social capital*. Oxford University Press.
- Chung, C. J., Lee, S., Barnett, G. A., & Kim, J. H. (2009). A comparative network analysis of the Korean Society of Journalism and Communication Studies (KSJCS) and the International Communication Association (ICA) in the era of hybridization. *Asian Journal of Communication*, 19(2), 170–191. <https://doi.org/10.1080/01292980902827003>
- Dimmick, J. W. (2003). *Media competition and coexistence: The theory of the niche*. Routledge.
- Doerfel, M. L., & Barnett, G. A. (1999). A semantic network analysis of the International Communication Association. *Human Communication Research*, 25(4), 589–603. <https://doi.org/10.1111/j.1468-2958.1999.tb00463.x>
- Doreian, P., & Fararo, T. J. (1985). Structural equivalence in a journal network. *Journal of the American Society for Information Science*, 36(1), 28–37. <https://doi.org/10.1002/asi.4630360103>
- Ehrenberg, A. S. C., Goodhardt, G. J., & Barwise, T. P. (1990). Double jeopardy revisited. *Journal of Marketing*, 54(3), 82–91. <https://doi.org/10.2307/1251818>
- Fletcher, R., & Nielsen, R. K. (2017). Are news audiences increasingly fragmented? A cross-national comparative analysis of cross-platform news audience fragmentation and duplication. *Journal of Communication*, 67(4), 476–498. <https://doi.org/10.1111/jcom.12315>
- Hannan, M. T., & Freeman, J. (1989). *Organizational ecology*. Harvard University Press.
- International Communication Association. (2021, October). *ICA bylaws*. <https://www.icaheadq.org/page/Bylaws>
- Jetten, J., Spears, R., & Manstead, A. S. R. (1998). Defining dimensions of distinctiveness: Group variability makes a difference to differentiation. *Journal of Personality and Social Psychology*, 74(6), 1481–1492. <https://doi.org/10.1037/0022-3514.74.6.1481>
- Krivitsky, P. N., & Handcock, M. S. (2014). A separable model for dynamic networks. *Journal of the Royal Statistical Society Series B, Statistical methodology*, 76(1), 29–46. <https://doi.org/10.1111/rssb.12014>
- Lee, S., & Lee, C. (2015). Creative interaction and multiplexity in intra-organizational networks. *Management Communication Quarterly*, 29(1), 56–83. <https://doi.org/10.1177/0893318914553120>
- Lee, S., & Monge, P. (2011). The coevolution of multiplex communication networks in organizational communities. *Journal of Communication*, 61(4), 758–779. <https://doi.org/10.1111/j.1460-2466.2011.01566.x>
- Leifeld, P., Cranmer, S. J., & Desmarais, B. A. (2018). Temporal exponential random graph models with btergm: Estimation and bootstrap confidence intervals. *Journal of Statistical Software*, 83(6), 1–36. <https://doi.org/10.18637/jss.v083.i06>

- Leonardi, P., & Contractor, N. (2018). Better people analytics: Measure who they know, not just who they are. *Harvard Business Review*, 96(6), 90–101.
- McPhee, W. N. (1963). *Formal theories of mass behavior*. The Free Press. <https://www.biblio.com/book/formal-theories-mass-behavior-mcphee-william/d/1237182311>.
- McPherson, M. (1983). An ecology of affiliation. *American Sociological Review*, 48(4), 519–532. <https://doi.org/10.2307/2117719>
- McPherson, M. (2004). A Blau space primer: Prolegomenon to an ecology of affiliation. *Industrial and Corporate Change*, 13(1), 263–280. <https://doi.org/10.1093/icc/13.1.263>
- McPherson, M., Smith-Lovin, L., & Cook, J. M. (2001). Birds of a feather: Homophily in social networks. *Annual Review of Sociology*, 27(1), 415–444. <https://doi.org/10.1146/annurev.soc.27.1.415>
- Monge, P., & Poole, M. S. (2008). The evolution of organizational communication. *Journal of Communication*, 58(4), 679–692. <https://doi.org/10.1111/j.1460-2466.2008.00408.x>
- Monge, P. R., & Contractor, N. (2003). *Theories of communication networks*. Oxford University Press.
- Mukerjee, S., Majó-Vázquez, S., & González-Bailón, S. (2018). Networks of audience overlap in the consumption of digital news. *Journal of Communication*, 68(1), 26–50. <https://doi.org/10.1093/joc/jqx007>
- Shumate, M., Pilny, A., Atouba, Y. C., Kim, J., Pena-y-Lillo, M., Cooper, K. R., Sahagun, A., & Yang, S. (2013). A taxonomy of communication networks. *Annals of the International Communication Association*, 37(1), 95–123. <https://doi.org/10.1080/23808985.2013.11679147>
- Song, H., Eberl, J. M., & Eisele, O. (2020). Less fragmented than we thought? Toward clarification of a subdisciplinary linkage in communication science, 2010–2019. *Journal of Communication*, 70(3), 310–334. <https://doi.org/10.1093/joc/jqaa009>
- Stinchcombe, A. L. (1965). Social structure and organizations. In J. P. March (Ed.), *Handbook of organizations* (pp. 142–193). Rand McNally.
- Tasselli, S., Zappa, P., & Lomi, A. (2020). Bridging cultural holes in organizations: The dynamic structure of social networks and organizational vocabularies within and across subunits. *Organization Science*, 31(5), 1292–1312. <https://doi.org/10.1287/orsc.2019.1352>
- Tenenboim-Weinblatt, K., & Lee, C. (2020). Speaking across communication subfields. *Journal of Communication*, 70(3), 303–309. <https://doi.org/10.1093/joc/jqaa012>
- Uzzi, B., Mukherjee, S., Stringer, M., & Jones, B. (2013). Atypical combinations and scientific impact. *Science*, 342(6157), 468–472. <https://doi.org/10.1126/science.1240474>
- Wageman, R., Gardner, H., & Mortensen, M. (2012). The changing ecology of teams: New directions for teams research. *Journal of Organizational Behavior*, 33(3), 301–315. <https://doi.org/10.1002/job.1775>
- Wang, X., Butler, B. S., & Ren, Y. (2013). The impact of membership overlap on growth: An ecological competition view of online groups. *Organization Science*, 24(2), 414–431. <https://doi.org/10.1287/orsc.1120.0756>
- Wasserman, S., & Faust, K. (1994). *Social network analysis: Methods and applications*. Cambridge University Press.
- Weber, M. S. (2017). Unseen disruptions and the emergence of new organizations. *Communication Theory*, 27(1), 92–113. <https://doi.org/10.1111/comt.12105>
- Webster, J. G. (2014). *The marketplace of attention: How audiences take shape in a digital age*. The MIT Press.
- Webster, J. G., & Ksiazek, T. B. (2012). The dynamics of audience fragmentation: Public attention in an age of digital media. *Journal of Communication*, 62(1), 39–56. <https://doi.org/10.1111/j.1460-2466.2011.01616.x>
- Wellman, B., Quan-Haase, A., Boase, J., Chen, W., Hampton, K., Díaz, I., & Miyata, K. (2003). The social affordances of the Internet for networked individualism. *Journal of Computer-Mediated Communication*, 8(3). <https://doi.org/10.1111/j.1083-6101.2003.tb00216.x>
- Woo, D., Myers, K. K. (2020). Organizational membership negotiation of boundary spanners: Becoming a competent jack of all trades and master of ... interactional expertise. *Management Communication Quarterly*, 34(1), 85–120. <https://doi.org/10.1177/0893318919887371>
- Xu, Y. (2022). Evolution of audience duplication networks among social networking sites: Exploring the influences of preferential attachment, audience size, and niche width. *New Media & Society*, 24(9), 2068–2087. <https://doi.org/10.1177/1461444821993048>
- Xu, Y., & Peng, T. Q. (2023). Ecological constraints on audience size in the digital media system: Evidence from the longitudinal tracking data from 2019 to 2022. *Human Communication Research*, 50(1), 14–26. <https://doi.org/10.1093/hcr/hqad028>
- Yang, A., & Taylor, M. (2015). Looking over, looking out, and moving forward: Positioning public relations in theorizing organizational network ecologies. *Communication Theory*, 25(1), 91–115. <https://doi.org/10.1111/comt.12049>
- Zajonc, R. B. (1968). Attitudinal effects of mere exposure. *Journal of Personality and Social Psychology*, 9(2, Pt.2), 1–27. <https://doi.org/10.1037/h0025848>